

环的直积

定义(环的直积) $(R_i)_{i \in I}$ 是一族环, 下标 i 遍历某个非空集 I . 把 $(R_i)_{i \in I}$ 看作一族集合, 则这一族集合的 Cartesian product 为:

~~$$\prod_{i \in I} R_i = \{ (a_i)_{i \in I} \mid \forall i \in I, a_i \in R_i \}$$~~

$$\prod_{i \in I} R_i = \{ (r_i)_{i \in I} \mid \forall i \in I, r_i \in R_i \}$$

在 $\prod_{i \in I} R_i$ 上定义加法运算: $\forall (r_i)_{i \in I} \in \prod_{i \in I} R_i, (r'_i)_{i \in I} \in \prod_{i \in I} R_i,$

$$(r_i)_{i \in I} + (r'_i)_{i \in I} := (r_i + r'_i)_{i \in I}$$

在 $\prod_{i \in I} R_i$ 上定义乘法运算: $\forall (r_i)_{i \in I} \in \prod_{i \in I} R_i, (r'_i)_{i \in I} \in \prod_{i \in I} R_i,$

$$(r_i)_{i \in I} \cdot (r'_i)_{i \in I} := (r_i \cdot r'_i)_{i \in I}$$

则有: $(\prod_{i \in I} R_i, +, \cdot, (0_i)_{i \in I}, (1_i)_{i \in I})$ 是环.

Proof: 首先证明加法运算和乘法运算良定义.

$\forall (r_i)_{i \in I} \in \prod_{i \in I} R_i, \forall (r'_i)_{i \in I} \in \prod_{i \in I} R_i,$ 有:

$\forall i \in I, r_i \in R_i, r'_i \in R_i. \because R_i \text{ 是环} \therefore r_i + r'_i \in R_i, r_i \cdot r'_i \in R_i$

$\therefore (r_i)_{i \in I} + (r'_i)_{i \in I} = (r_i + r'_i)_{i \in I} \in \prod_{i \in I} R_i$

$(r_i)_{i \in I} \cdot (r'_i)_{i \in I} = (r_i \cdot r'_i)_{i \in I} \in \prod_{i \in I} R_i$

$\therefore \prod_{i \in I} R_i$ 上的加法运算和乘法运算良定义.

$\forall (r_i)_{i \in I}, (r'_i)_{i \in I}, (r''_i)_{i \in I} \in \prod_{i \in I} R_i,$ 有:

$$\begin{aligned} ((r_i)_{i \in I} + (r'_i)_{i \in I}) + (r''_i)_{i \in I} &= (r_i + r'_i)_{i \in I} + (r''_i)_{i \in I} = ((r_i + r'_i) + r''_i)_{i \in I} \\ &= (r_i + (r'_i + r''_i))_{i \in I} = (r_i)_{i \in I} + (r'_i + r''_i)_{i \in I} = (r_i)_{i \in I} + ((r'_i)_{i \in I} + (r''_i)_{i \in I}) \end{aligned}$$

对 $\forall (r_i)_{i \in I} \in \prod_{i \in I} R_i$, 有:

$$(r_i)_{i \in I} + (0_i)_{i \in I} = (r_i + 0_i)_{i \in I} = (r_i)_{i \in I}$$

$$(0_i)_{i \in I} + (r_i)_{i \in I} = (0_i + r_i)_{i \in I} = (r_i)_{i \in I}$$

$$\therefore (r_i)_{i \in I} + (0_i)_{i \in I} = (r_i)_{i \in I} = (0_i)_{i \in I} + (r_i)_{i \in I} \quad \therefore \text{零元性质得证.}$$

对 $\forall (r_i)_{i \in I}, (r'_i)_{i \in I} \in \prod_{i \in I} R_i$, 有:

$$(r_i)_{i \in I} + (r'_i)_{i \in I} = (r_i + r'_i)_{i \in I} = (r'_i + r_i)_{i \in I} = (r'_i)_{i \in I} + (r_i)_{i \in I}$$

$\therefore$ 加法交换律得证.

对 $\forall (r_i)_{i \in I} \in \prod_{i \in I} R_i$, 有: 对 $\forall i \in I, r_i \in R_i \quad \therefore \exists -r_i \in R_i, \text{ s.t.}$

$$r_i + (-r_i) = 0_i$$

$\therefore (-r_i)_{i \in I} \in \prod_{i \in I} R_i$, 且有:

$$(r_i)_{i \in I} + (-r_i)_{i \in I} = (r_i + (-r_i))_{i \in I} = (0_i)_{i \in I}$$

$$\therefore -(r_i)_{i \in I} = (-r_i)_{i \in I}$$

对 $\forall (r_i)_{i \in I} \in \prod_{i \in I} R_i$, 有:

$$(r_i)_{i \in I} \cdot (1_i)_{i \in I} = (r_i \cdot 1_i)_{i \in I} = (r_i)_{i \in I}$$

$$(1_i)_{i \in I} \cdot (r_i)_{i \in I} = (1_i \cdot r_i)_{i \in I} = (r_i)_{i \in I} \quad \therefore \text{幺元性质得证.}$$

又 $\forall (r_i)_{i \in I}, (r'_i)_{i \in I}, (r''_i)_{i \in I} \in \prod_{i \in I} R_i$, 有:

$$\left((r_i)_{i \in I} \cdot (r'_i)_{i \in I} \right) \cdot (r''_i)_{i \in I} = (r_i \cdot r'_i)_{i \in I} \cdot (r''_i)_{i \in I} = ((r_i \cdot r'_i) \cdot r''_i)_{i \in I}$$

$$= (r_i \cdot (r'_i \cdot r''_i))_{i \in I} = (r_i)_{i \in I} \cdot (r'_i \cdot r''_i)_{i \in I} = (r_i)_{i \in I} \cdot \left((r'_i)_{i \in I} \cdot (r''_i)_{i \in I} \right)$$

$$\left((r_i)_{i \in I} + (r'_i)_{i \in I} \right) \cdot (r''_i)_{i \in I} = (r_i + r'_i)_{i \in I} \cdot (r''_i)_{i \in I} = ((r_i + r'_i) \cdot r''_i)_{i \in I}$$

$$= (r_i \cdot r''_i + r'_i \cdot r''_i)_{i \in I} = (r_i \cdot r''_i)_{i \in I} + (r'_i \cdot r''_i)_{i \in I} = (r_i)_{i \in I} \cdot (r''_i)_{i \in I} + (r'_i)_{i \in I} \cdot (r''_i)_{i \in I}$$

$$(r''_i)_{i \in I} \cdot \left((r_i)_{i \in I} + (r'_i)_{i \in I} \right) = (r''_i)_{i \in I} \cdot (r_i + r'_i)_{i \in I} = (r''_i \cdot (r_i + r'_i))_{i \in I}$$

$$= (r''_i \cdot r_i + r''_i \cdot r'_i)_{i \in I} = (r''_i \cdot r_i)_{i \in I} + (r''_i \cdot r'_i)_{i \in I} = (r''_i)_{i \in I} \cdot (r_i)_{i \in I} + (r''_i)_{i \in I} \cdot (r'_i)_{i \in I}$$

$\therefore \left(\prod_{i \in I} R_i, +, \cdot, (0_i)_{i \in I}, (1_i)_{i \in I} \right)$ 是环. $\square$

例: 又 $\forall i \in I, R_i = R$. 此时有:

$$\prod_{i \in I} R_i = \left\{ \text{映射 } f: I \rightarrow R \mid \text{又 } \forall i \in I, f(i) \in R \right\}$$

$$= \left\{ \text{映射 } f: I \rightarrow R \right\} = R^I$$

$$\text{又 } \forall f, g \in R^I, \text{ 有: } \begin{array}{l} f: I \longrightarrow R \\ i \longmapsto f(i) \end{array}, \quad \begin{array}{l} g: I \longrightarrow R \\ i \longmapsto g(i) \end{array}$$

$$\begin{array}{l} f+g: I \longrightarrow R \\ i \longmapsto (f+g)(i) = f(i) + g(i) \end{array}$$

$\downarrow$
环R中的加法

$$\begin{array}{l} f \cdot g: I \longrightarrow R \\ i \longmapsto (f \cdot g)(i) = f(i) \cdot g(i) \end{array}$$

$\downarrow$
环R中的乘法

环 R^I 的加法零元 $0_{R^I} : I \longrightarrow R$ 是常值映射
 $i \longmapsto 0_R$

环 R^I 的乘法幺元 $1_{R^I} : I \longrightarrow R$ 是常值映射
 $i \longmapsto 1_R$

$(R^I, +, \cdot, 0_{R^I}, 1_{R^I})$ 是环